

A disproof of Conjecture 00000007785

The cube has Mahler product $4^n/n! < 4^n$

1 The conjecture

Conjecture 00000007785 reads (English version):

Definition: The Mahler product $M(K) = \text{vol}(K) \text{vol}(K^\circ)$ with polar K° ; the Mahler conjecture asserts $M(K) \geq 4^n$ in the symmetric case. **Conjecture:** The symmetric n -dimensional extremal bodies are the Hanner polytopes (equality cases), with strict inequality outside finite products of the Hanner family and finitely many low-dimensional exceptions ($n \leq 3$); the nonsymmetric case $M(K) \geq (n+1)^{n+1}/(n!)^2$ is attained uniquely by the simplex; and both statements have stability versions: $M(K) \leq 4^n(1+\varepsilon)$ implies K is $O(\varepsilon^{1/2})$ close to a Hanner body.

The Chinese version states the same formulas: the symmetric bound $M(K) \geq 4^n$, the Hanner polytopes as the equality cases, and the stability hypothesis $M(K) \leq 4^n(1+\varepsilon)$.

Reading. We read the text literally. Its symmetric clause contains two claims about origin-symmetric convex bodies $K \subset \mathbb{R}^n$:

- (a) the inequality $M(K) \geq 4^n$;
- (b) the Hanner polytopes are the equality cases of this bound, so every Hanner polytope satisfies $M(K) = 4^n$.

We refute both (a) and (b) in every dimension $n \geq 2$. The conjecture is a conjunction, so it is false. We do not address the nonsymmetric clause $(n+1)^{n+1}/(n!)^2$ or the stability clause, and we make no claim about the statement obtained by replacing 4^n with $4^n/n!$ (which is the classical Mahler conjecture; see Remark 9).

2 Definitions and conventions

We work in $\mathbb{R}^n$, modelled in Lean as `Fin n → ℝ`, with the standard dot product $x \cdot y = \sum_i x_i y_i$ and Lebesgue measure `volume` (the product measure, which is Euclidean volume). Volumes take values in $[0, \infty]$.

Definition 1. A *convex body* is a compact convex set with non-empty interior. Wikipedia (Mahler volume): “A convex body in Euclidean space is defined as a compact convex set with non-empty interior.” It is *origin-symmetric* if $-x \in K$ whenever $x \in K$. (Lean: `IsSymmetricConvexBody`.)

Definition 2. The polar of K is $K^\circ = \{y : x \cdot y \leq 1 \text{ for all } x \in K\}$, and the Mahler product is $M(K) = \text{vol}(K) \text{vol}(K^\circ)$. (Lean: `polar`, `mahler`.) This is the definition in the Wikipedia article “Mahler volume”.

Definition 3. The Hanner polytopes are generated recursively. Wikipedia (Hanner polytope) gives the rules “A line segment is a one-dimensional Hanner polytope.”, “The Cartesian product of every two Hanner polytopes is another Hanner polytope, whose dimension is the sum of the dimensions of the two given polytopes.” and “The dual of a Hanner polytope is another Hanner polytope of the same dimension.” The Lean predicate `IsHanner n K` is the inductive closure of the segment $[-1, 1] \subset \mathbb{R}^1$ under Cartesian products (coordinates concatenated by `Fin.castAdd/Fin.natAdd`) and polar duals. It uses only the symmetric segment $[-1, 1]$, so it describes a subclass of the Hanner polytopes. Showing that a body lies in this subclass therefore shows that it is a Hanner polytope.

The cube is $C_n = [-1, 1]^n = \{x : |x_i| \leq 1 \ \forall i\}$ and the cross-polytope is $Q_n = \{y : \sum_i |y_i| \leq 1\}$.

3 The proof

Lemma 4 (`polar_cube`). $C_n^\circ = Q_n$.

Proof. If $y \in Q_n$ and $x \in C_n$, then $x \cdot y \leq \sum_i |x_i| |y_i| \leq \sum_i |y_i| \leq 1$. Conversely, if $y \in C_n^\circ$, test with $x_i = 1$ when $y_i \geq 0$ and $x_i = -1$ otherwise. Then $x \in C_n$ and $x \cdot y = \sum_i |y_i| \leq 1$. $\square$

Lemma 5 (`volume_cube`, `volume_crossPolytope`). $\text{vol}(C_n) = 2^n$, and for $n \geq 1$, $\text{vol}(Q_n) = 2^n/n!$.

Proof. C_n is the box $\prod_i [-1, 1]$, so its volume is 2^n (`Mathlib.Real.volume_Icc_pi`). For Q_n , `Mathlib's volume_sum_rpow_le` with $p = 1$ and $r = 1$ gives $\text{vol}\{x : \sum_i |x_i| \leq 1\} = (2\Gamma(2))^n / \Gamma(n+1) = 2^n/n!$, using $\Gamma(2) = 1$ and $\Gamma(n+1) = n!$. $\square$

Lemma 6 (`mahler_cube`, `mahler_cube_lt`). For $n \geq 1$, $M(C_n) = 4^n/n!$. For $n \geq 2$, $M(C_n) < 4^n$.

Proof. $M(C_n) = 2^n \cdot 2^n/n! = 4^n/n!$, and $n! > 1$ when $n \geq 2$. $\square$

Lemma 7 (`cube_isSymmetricConvexBody`, `cube_isHanner`). C_n is an origin-symmetric convex body. For $n \geq 1$ it is a Hanner polytope.

Proof. C_n is a closed box, so it is compact and convex. It is symmetric, and it contains the open sup-norm unit ball, so 0 is an interior point. For the Hanner property, $C_1 = [-1, 1]$ is the segment, and C_{m+1} is the Cartesian product of C_m and C_1 (`Fin.forall_fin_add`). Induction finishes the proof. $\square$

Theorem 8 (`C7785.conjecture_7785_symmetric_clause_false`). For every $n \geq 2$: the cube C_n is a Hanner polytope and an origin-symmetric convex body with $M(C_n) = 4^n/n! < 4^n$. Hence

(i) it is false that $M(K) \geq 4^n$ for every origin-symmetric convex body $K \subset \mathbb{R}^n$;

(ii) it is false that every Hanner polytope in $\mathbb{R}^n$ satisfies $M(K) = 4^n$.

Claim (ii) shows that the Hanner polytopes are not the equality cases of the bound in (a), so claim (b) fails. Both failures occur in every dimension $n \geq 2$, so the clause “finitely many low-dimensional exceptions ($n \leq 3$)” cannot rescue the statement. That clause modifies only the strict-inequality part, and in any case the failure happens for all $n \geq 2$. In dimension 1 the bound reads $M([-1, 1]) = 4 = 4^1$, so there is no failure there.

Remark 9. The Wikipedia article “Mahler volume” states: “In particular, the polar body of a cube or hypercube is an octahedron or cross polytope.” It gives the Mahler volume of the hypercube as $4^n/\Gamma(n+1)$, which agrees with the lemma above. It also states: “The shapes with the minimum known Mahler volume are hypercubes, cross polytopes, and more generally the Hanner polytopes which include these two types of shapes, as well as their affine transformations.” The classical conjecture therefore has the constant $4^n/n!$, not the 4^n written in the statement. The conjecture under review writes 4^n in both languages, both in the bound and in the stability hypothesis $M(K) \leq 4^n(1+\varepsilon)$. We refute that literal statement and claim nothing about the corrected one, which remains open for $n \geq 4$.

4 Lean formalization and axiom audit

File `lean/Conjecture7785/Basic.lean` (namespace `C7785`, 156 lines, only `import Mathlib`, Lean/Mathlib v4.33.1). Main theorem:

```
theorem conjecture_7785_symmetric_clause_false (n : N) (hn : 2 <= n) :
  (IsHanner n (cube n) /\ IsSymmetricConvexBody (cube n) /\
    mahler (cube n) = ENNReal.ofReal (4 ^ n / n !)) /\ mahler (cube n) < 4 ^ n /\
    ~ (forall K, IsSymmetricConvexBody K -> (4 : ENNReal) ^ n <= mahler K) /\
    ~ (forall K, IsHanner n K -> mahler K = 4 ^ n)
```

lake build succeeds, and `#print axioms C7785.conjecture_7785_symmetric_clause_false` reports only `[propext, Classical.choice, Quot.sound]`. The file contains no `sorry` and no `native_decide`.